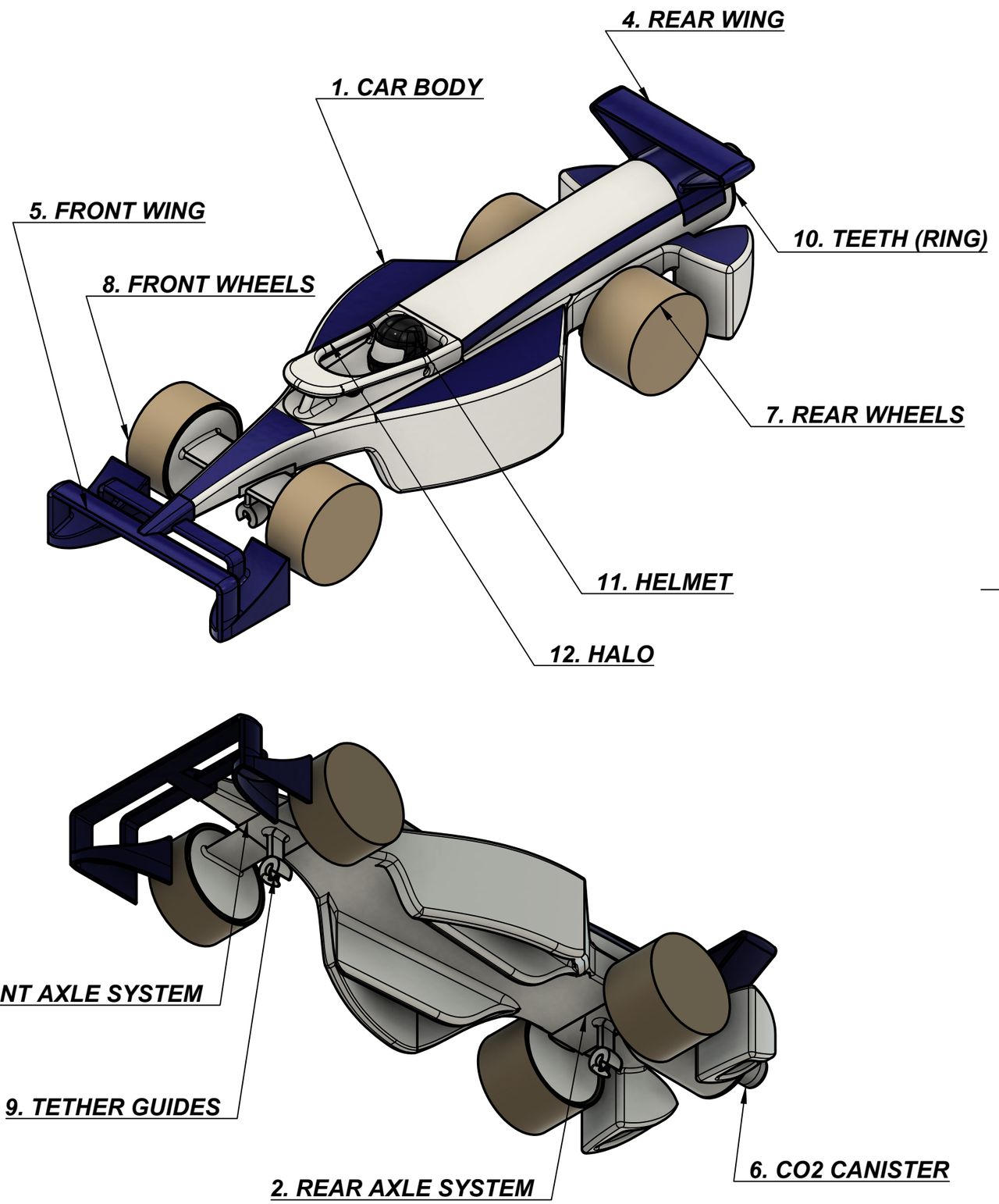
Maverick Motorsports
Engineering Drawings

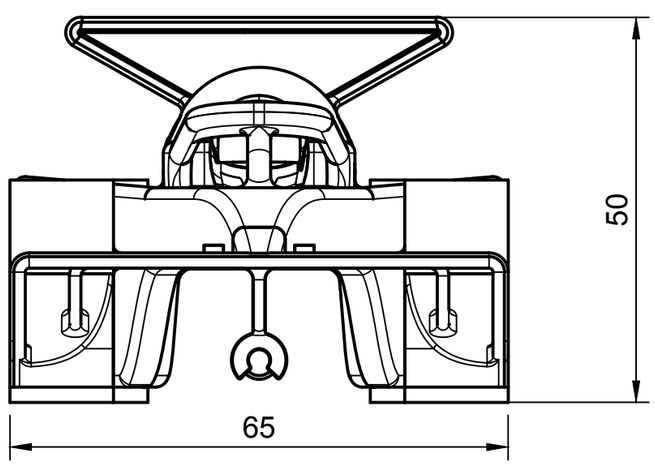
Part Number	Parts Name	Quantity	Description	Material
12	Halo	1	3D printed using SLS Technology	Nylon
11	Helmet	1	3D Printed using FDM Technology	Carbon Fibre
10	Teeth (Ring)	1	3D Printed using FDM Technology	Carbon Fibre
9	Tether Guides	2	Extracted from Fuji fishing guides	Silica Carbide
8	Front Wheels	2	VMC Machined	Ketron PEEK
7	Rear Wheels	2	VMC machined	Ketron PEEK
6	CO2 Canister	1	Provided by Stemplify	Aluminium
5	Front Wing	1	3D printed using FDM Technology	Carbon Fibre
4	Rear Wing	1	3D printed using FDM Technology	Carbon Fibre
3	Front Axle System	1	3D printed using SLS Technology	Nylon 12
2	Rear Axle System	1	3D printed using SLS Technology	Nylon 12
1	Car Body	1	CNC machined from Stemplify	Polyurethane Foam

Bill of Materials



IMPORTANT REGULATIONS
(Assembled car body, Power unit cartridge)

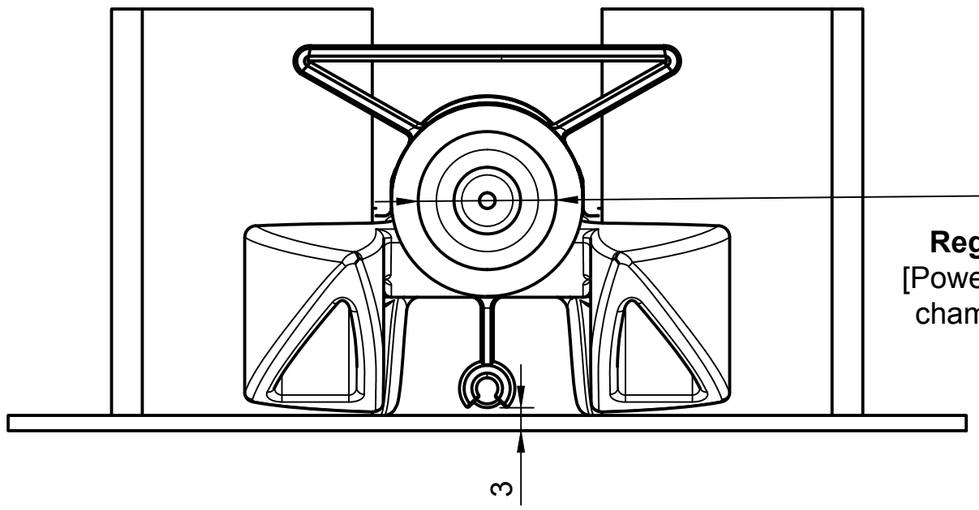
Front View



Regulation T3.5 (max. 65mm)
[Total height]

Regulation T3.4 (min. 65mm)
[Total width]

Rear View

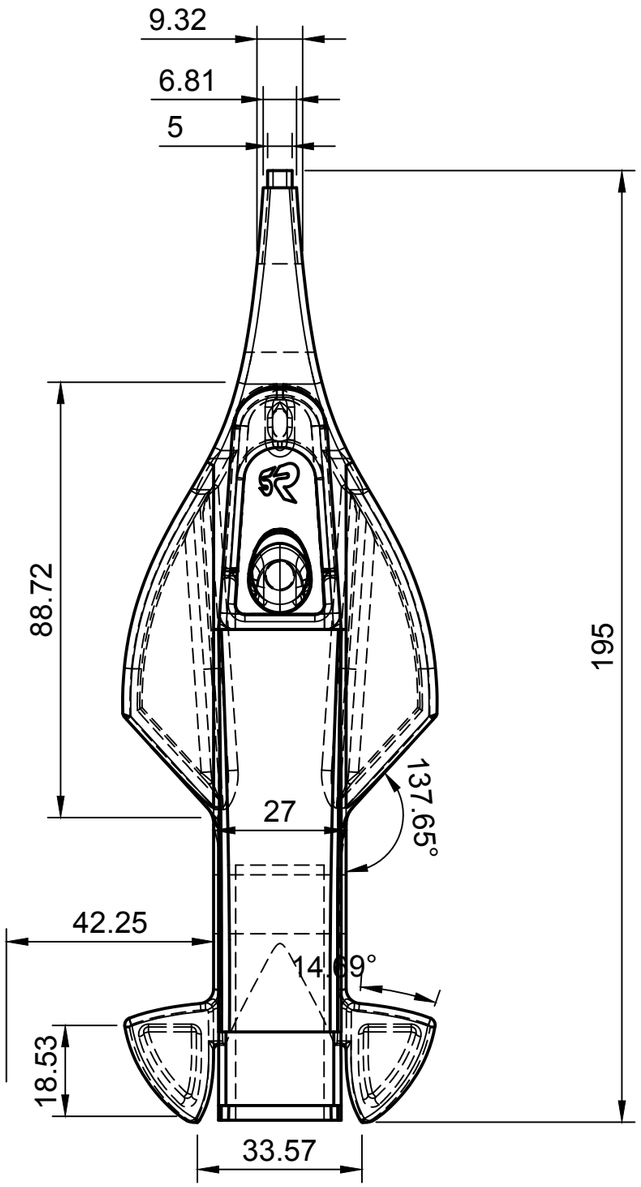


Regulation T5.1
[Power unit cartridge chamber diameter]

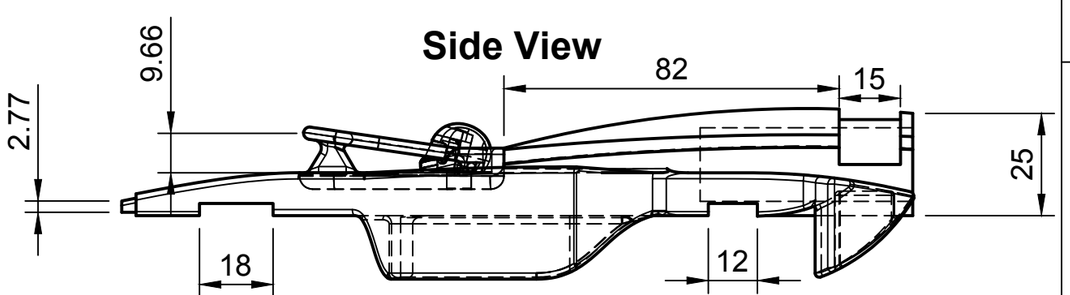
Regulation T3.7 (min. 1.5mm)
[Track clearance]

IMPORTANT DIMENSIONS FOR MANUFACTURING

Top View



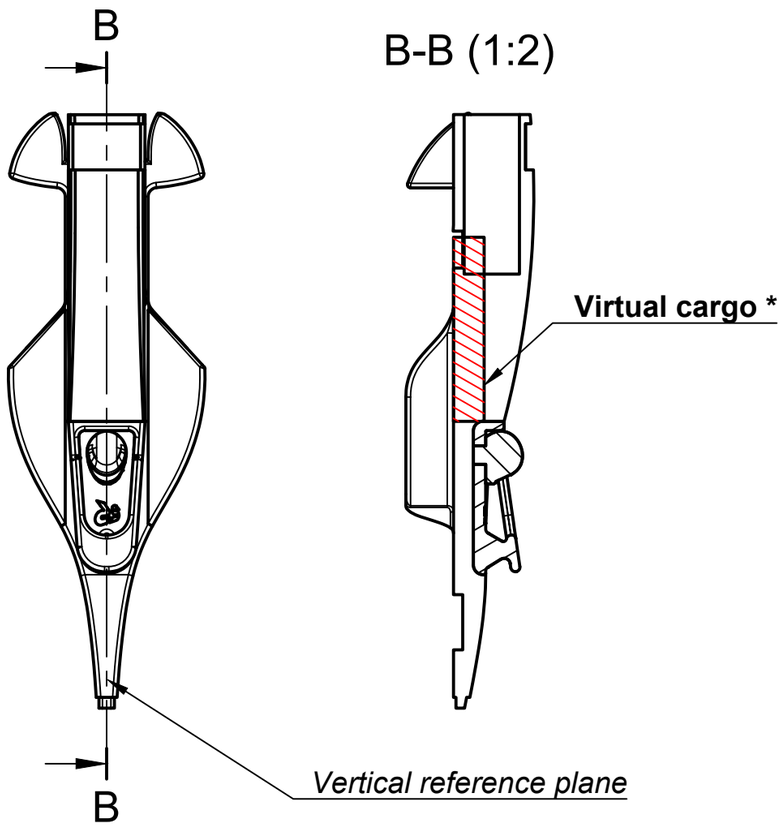
Side View



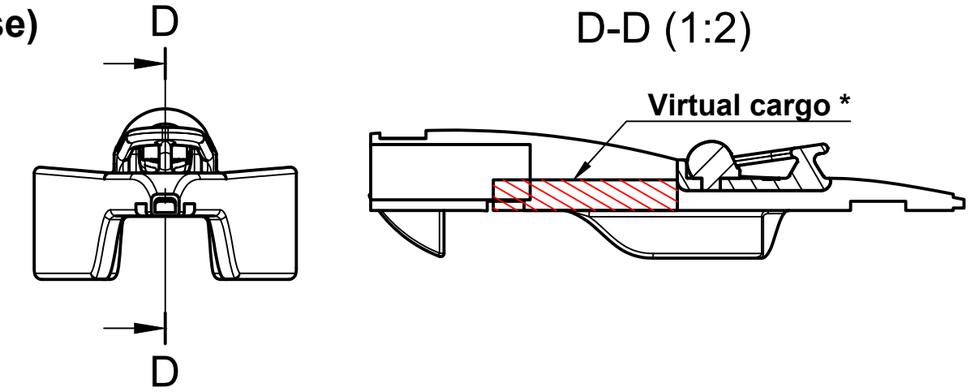
IMPORTANT REGULATIONS
(Virtual cargo, Wheelbase)

*** Regulation T4.3**
(Virtual cargo identification)

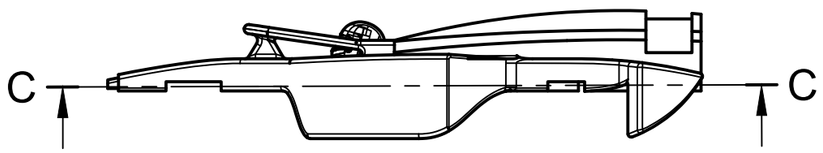
Section View (from Top)

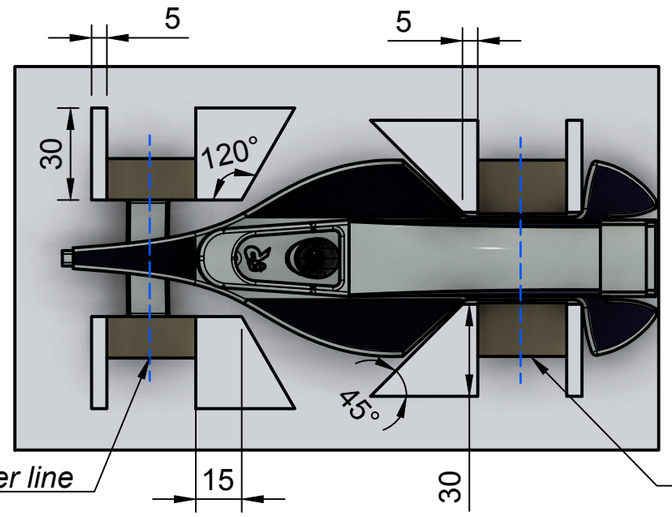


Section View (from Front)

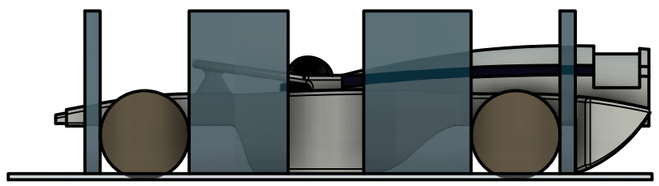


Section View (from Side)

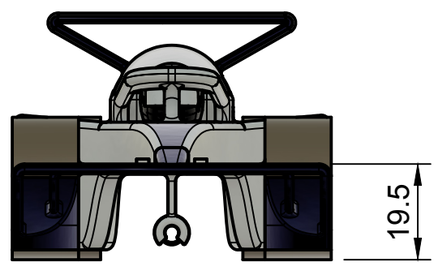




Regulation T7.9
[Wheel visibility in top and bottom views]



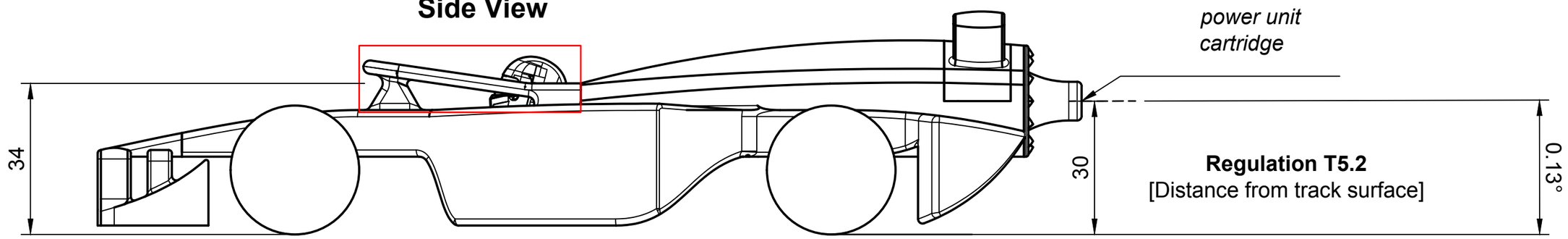
Regulation T7.10
[Wheel visibility in side view]



Regulation T7.11 (<20mm)
[Wheel visibility in front view]

IMPORTANT REGULATIONS
(Halo, Power unit cartridge, Wheel, Tether line guide)

Side View



Regulation T4.4.4
[Halo circular notch height]

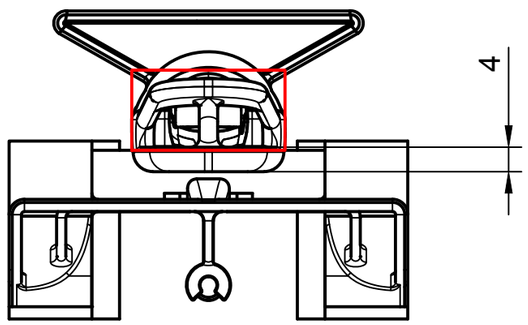
Regulation T4.4.2
[Halo visibility side view]

Regulation T5.2
[Distance from track surface]

Regulation T5.6 (min. 5mm)
[Cartridge visibility]

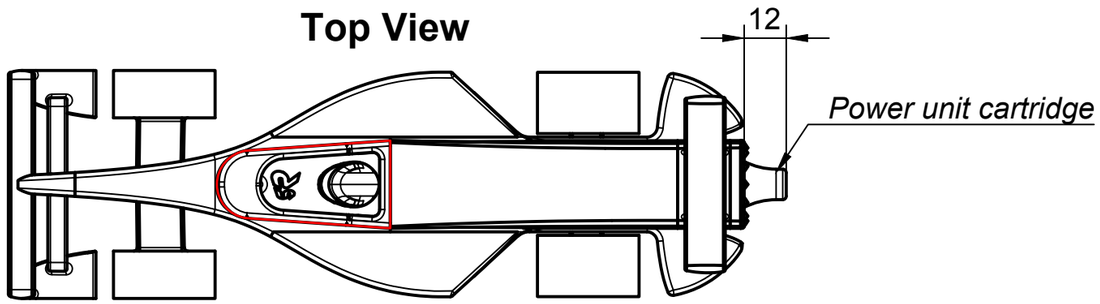
Regulation T5.4
[Max angle of chamber]

Front View



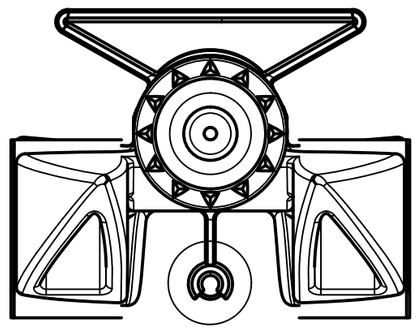
Regulation T4.4.2
[Halo visibility front view]

Top View



Regulation T4.4.3
[Halo visibility top view]

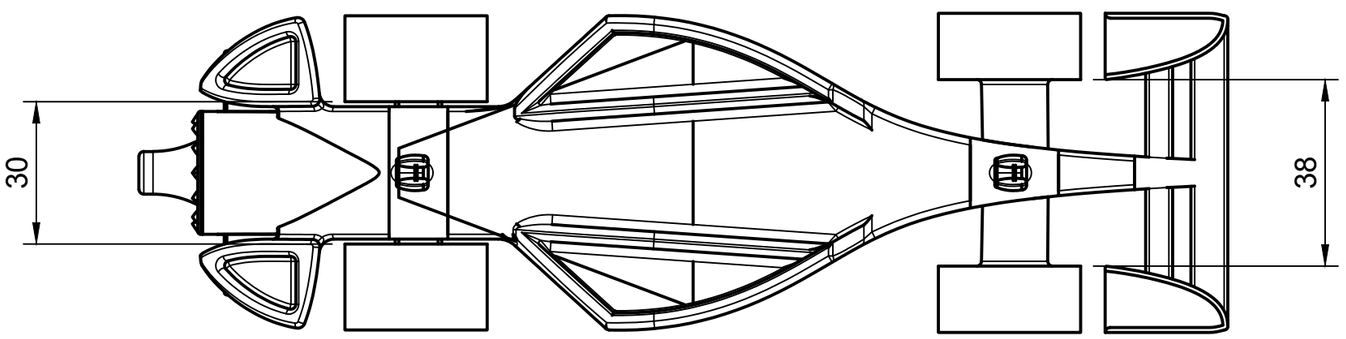
Rear View



A

Bottom View

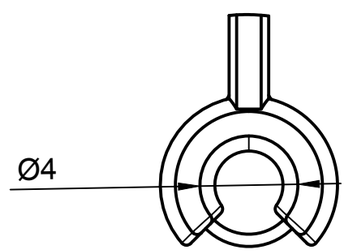
Regulation T7.2.2 (min. 30mm)
[Rear Wheel Distance]



Regulation T7.2.1 (min. 38mm)
[Front Wheel Distance]

Expanded View (A)

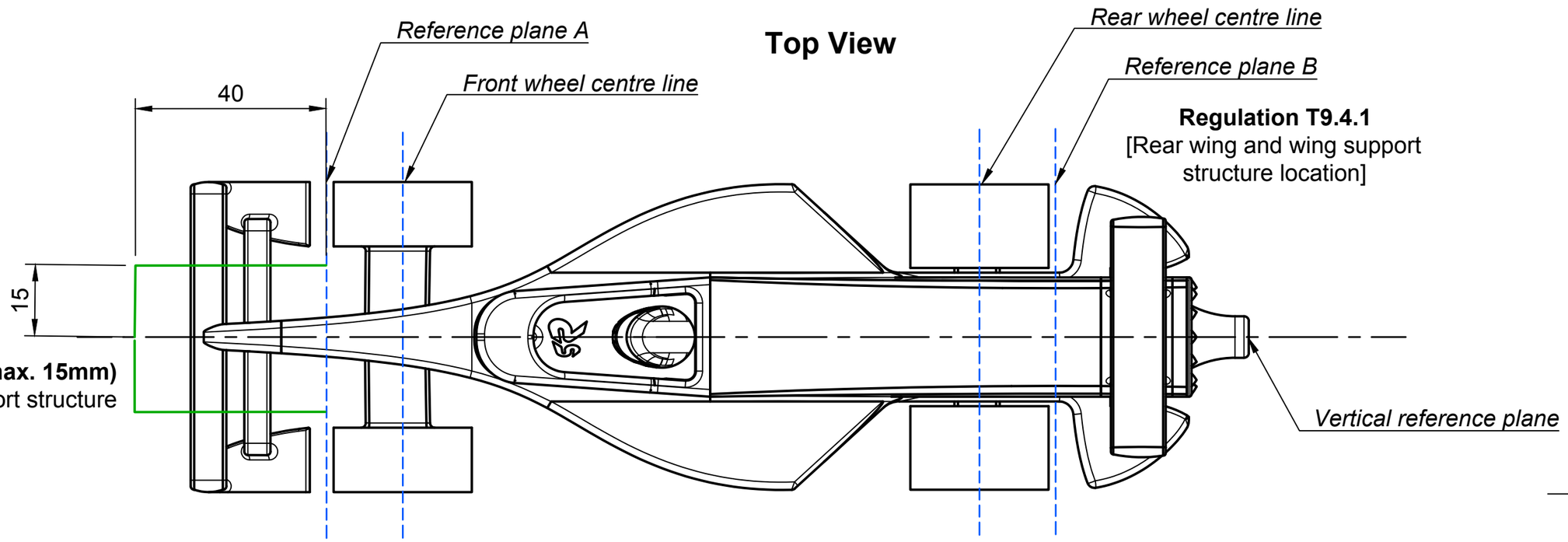
A (4:1)



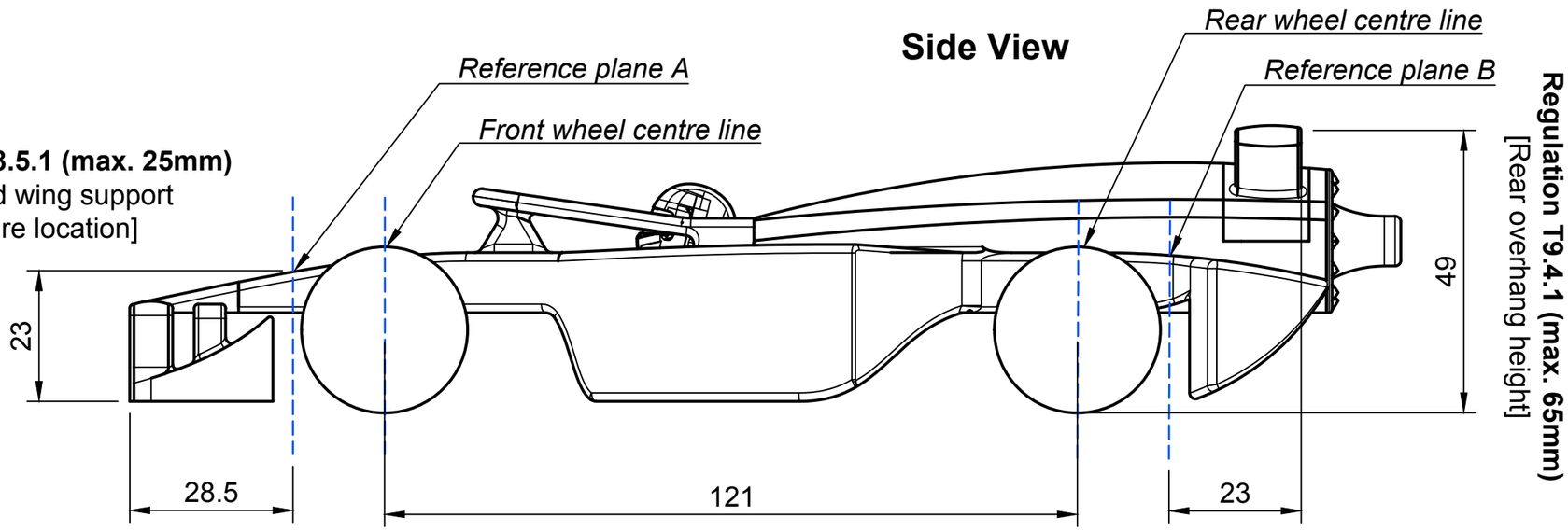
Regulation T6.2
[Tether line guide internal diameter]

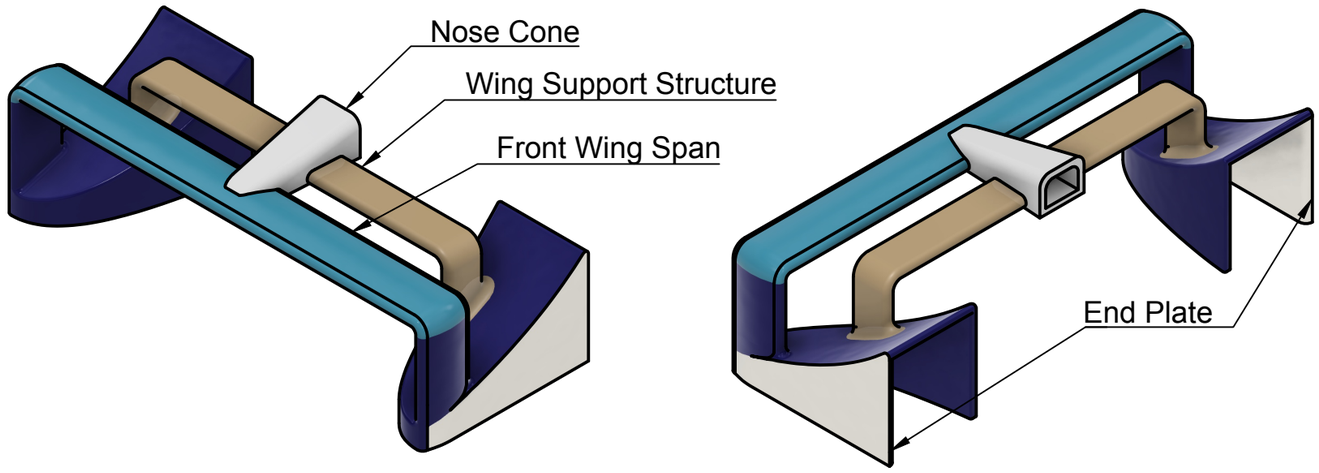
IMPORTANT REGULATIONS
(Front wing, Rear wing)

Top View



Side View



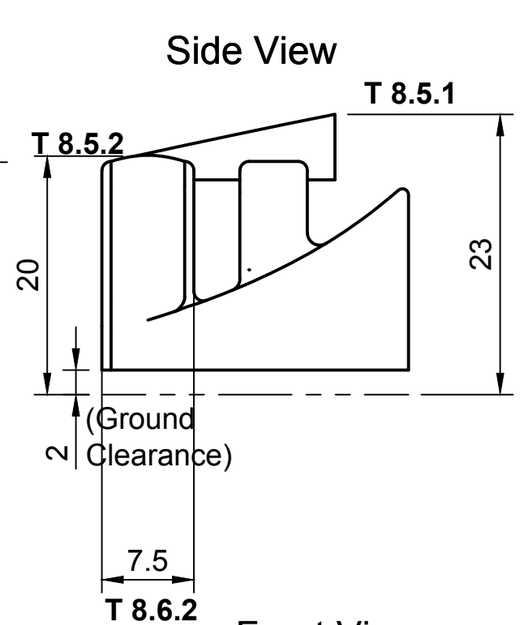


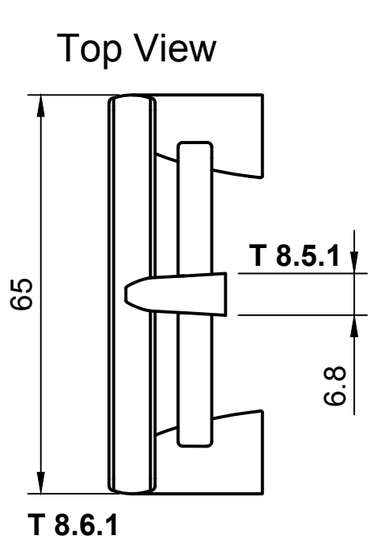
Isometric View of Front Wing Identifying Components (T 8.1)

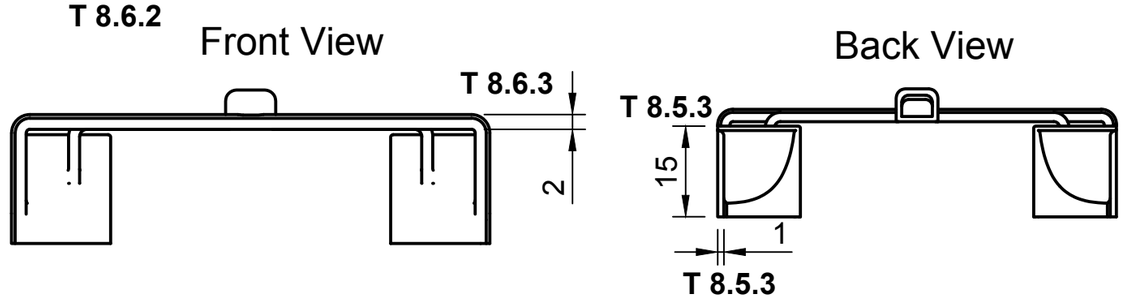
FRONT WING

Material: Carbon Fibre

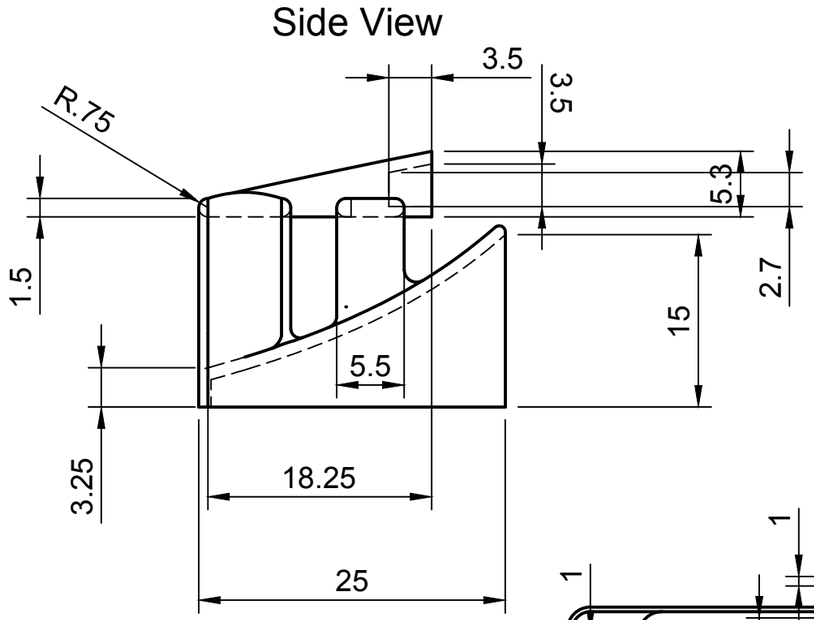
Chosen due to high strength to weight ratio

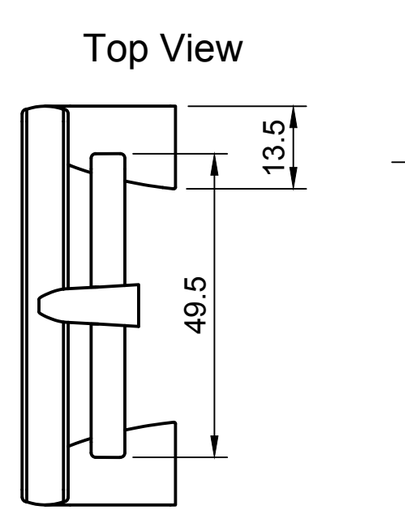


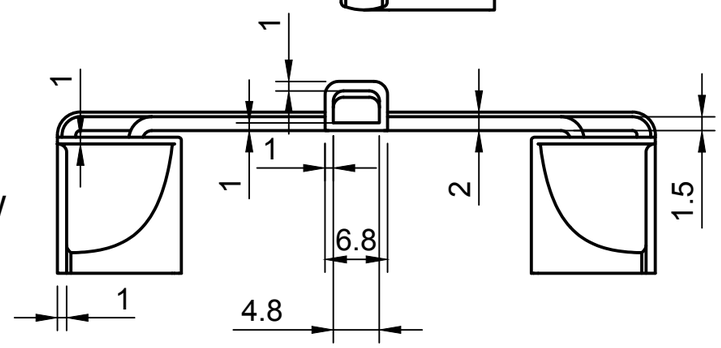




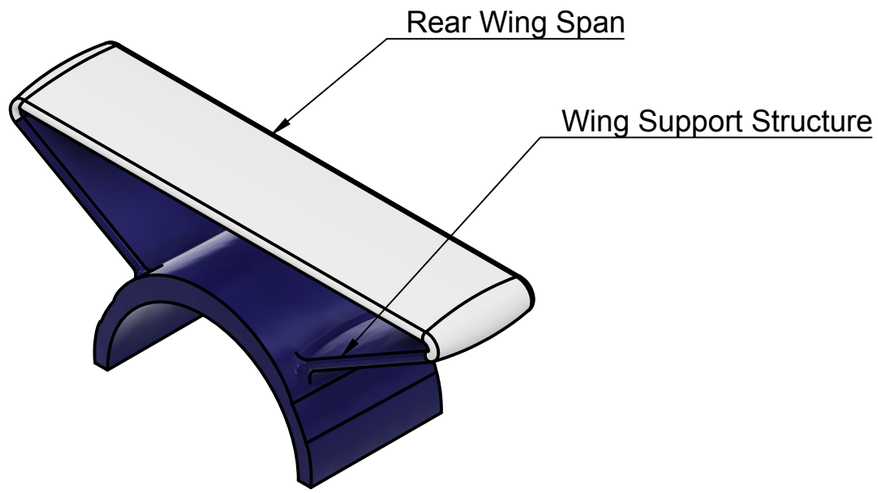
Important Regulations Regarding Front Wing







Important Dimensions For Manufacturing

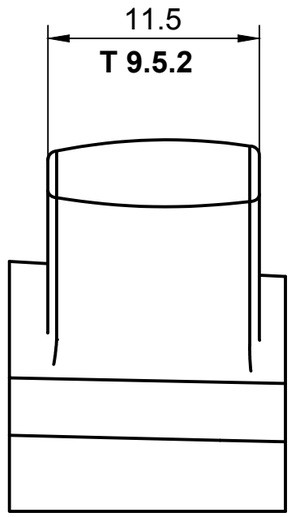


Isometric View of Rear Wing, Identifying Componenets (T 9.1)

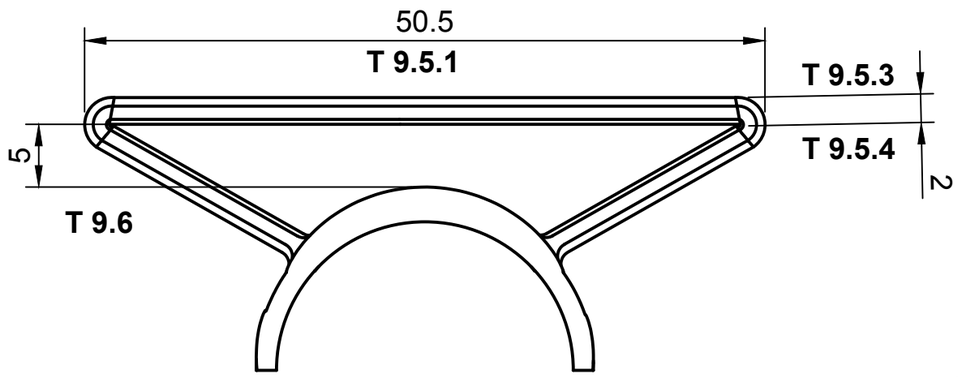
REAR WING

Material: Carbon Fibre

Chosen due to High Stength to Weight Ratio

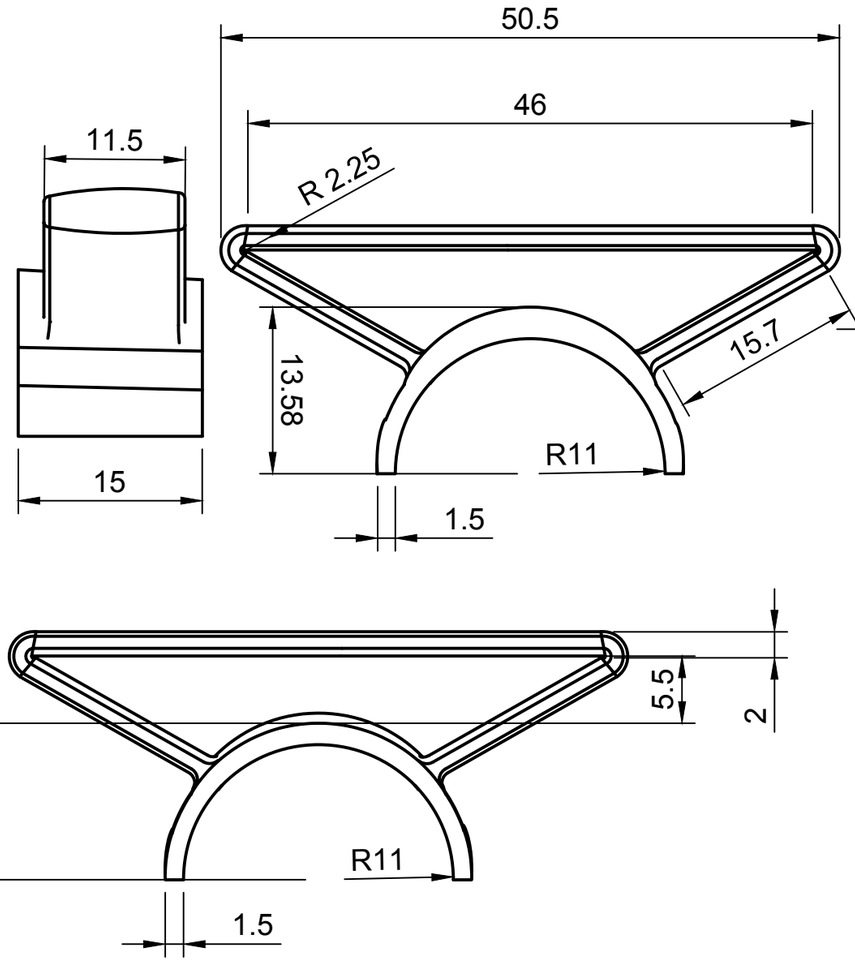


Side View

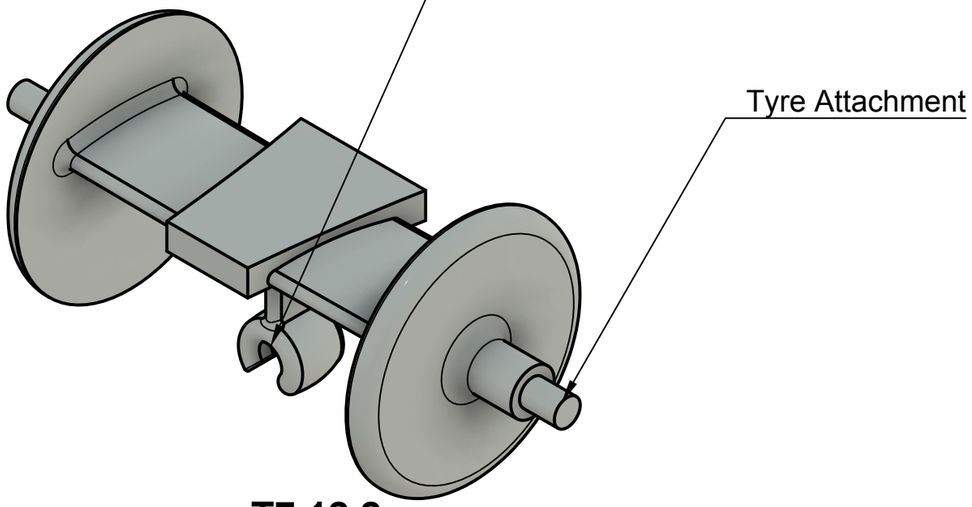


Front View

Imporant Regulations Related to Rear Wing



Imporant DImensions Related to Manufacturing

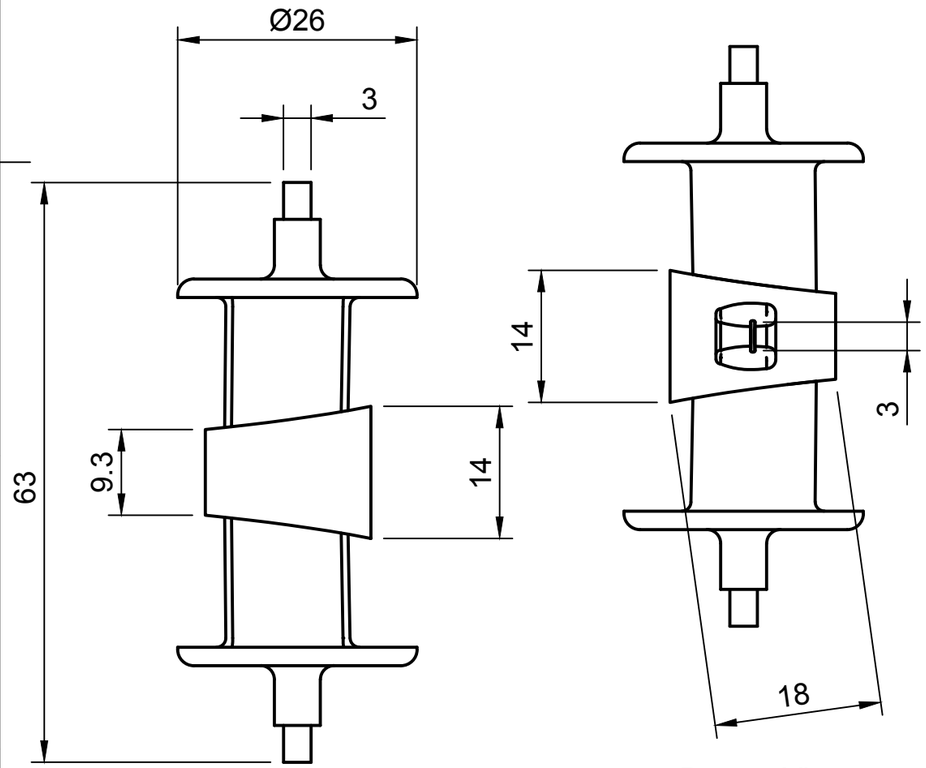


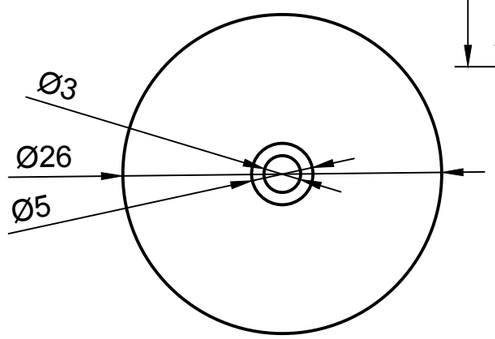
T7.12.2
Wheel support system identification

FRONT AXLE SYSTEM

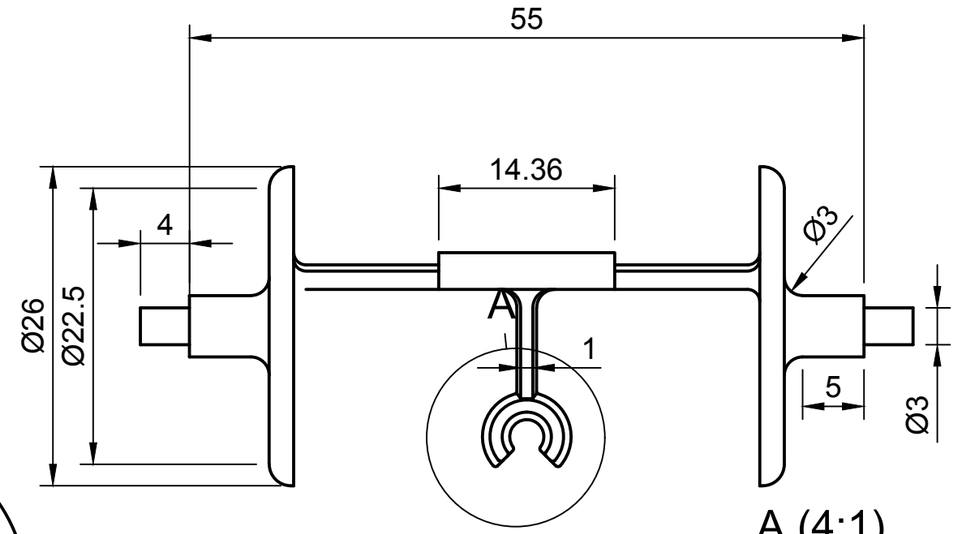
Material: Nylon 12

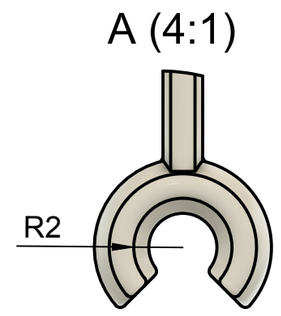
Chosen for its flexibility, lightweight and ease of manufacturing





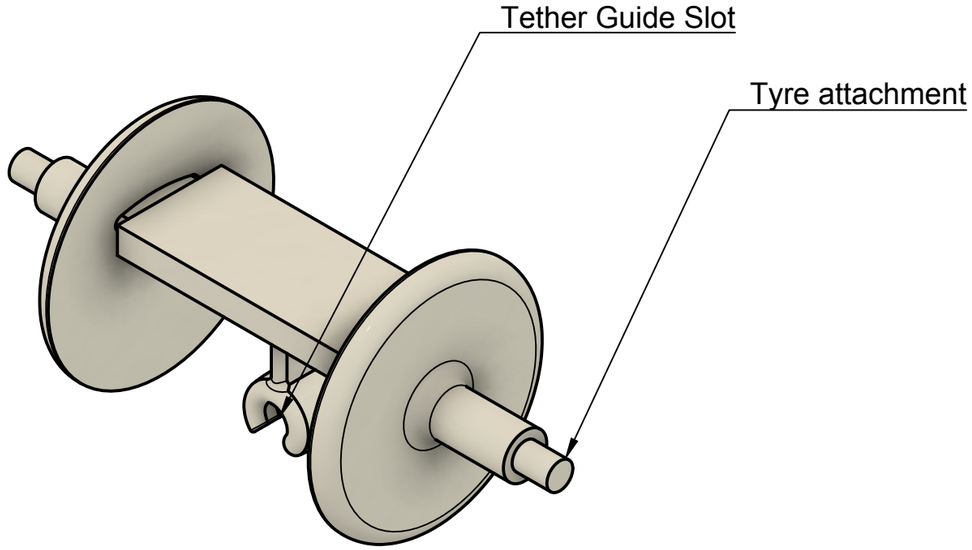
(radius of wheel is 29mm, Therefore axle system fits between 2 opposing wheels following the cylindrical volume condition)
T7.12





Important Regulations Regarding Axle System

Important Dimensions For Manufacturing

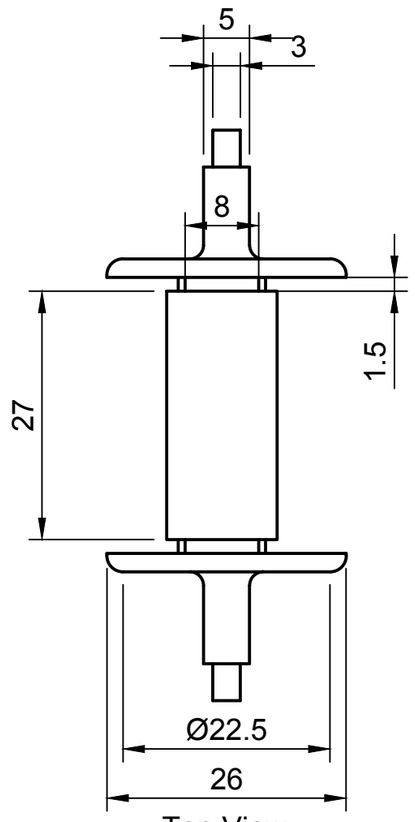


T7.12.2
Wheel support system identification

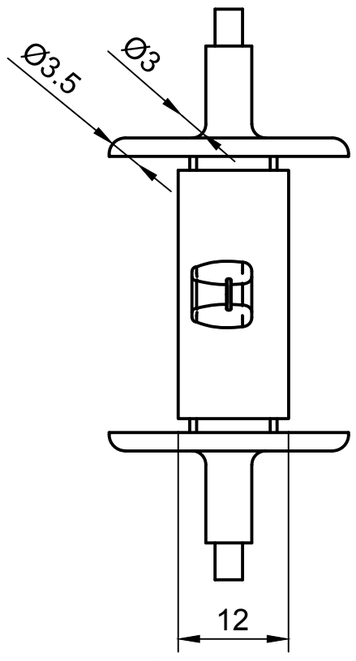
REAR AXLE SYSTEM

Material: Nylon 12

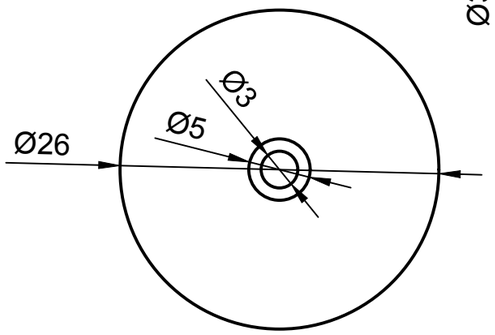
Chosen for its flexibility, lightweight and ease of manufacturing



Top View

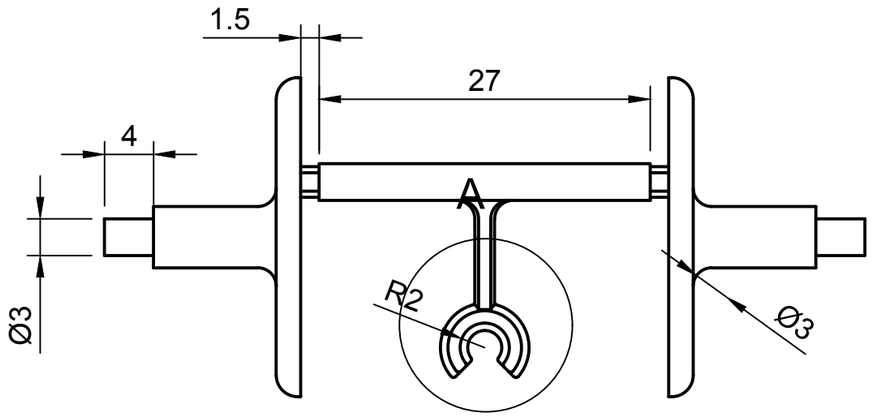


Bottom View

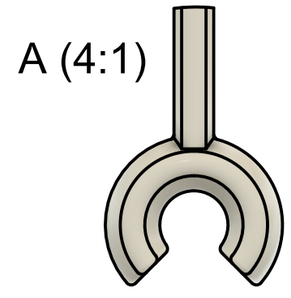


Side View
(radius of wheel is 29mm, Therefore axle system fits between 2 opposing wheels following the cylindrical volume condition)

T7.12



Front View



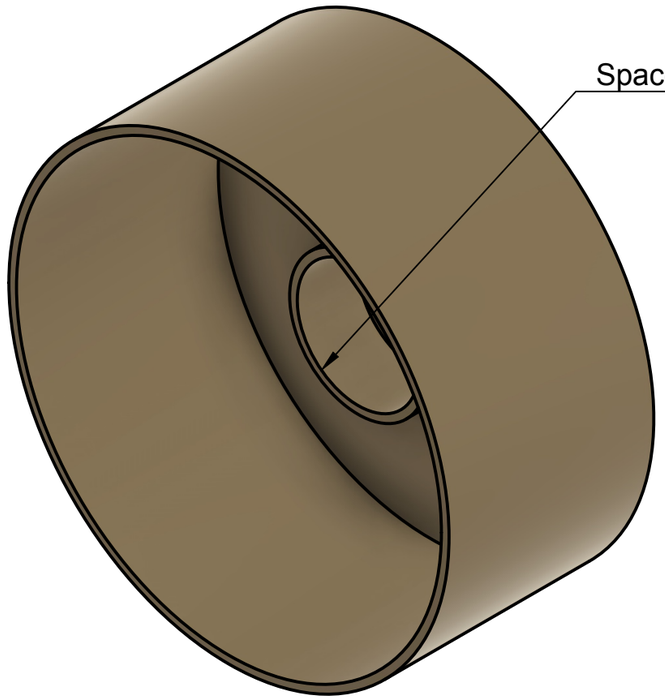
Important Regulations Regarding Axle System

Important Dimensions For Manufacturing

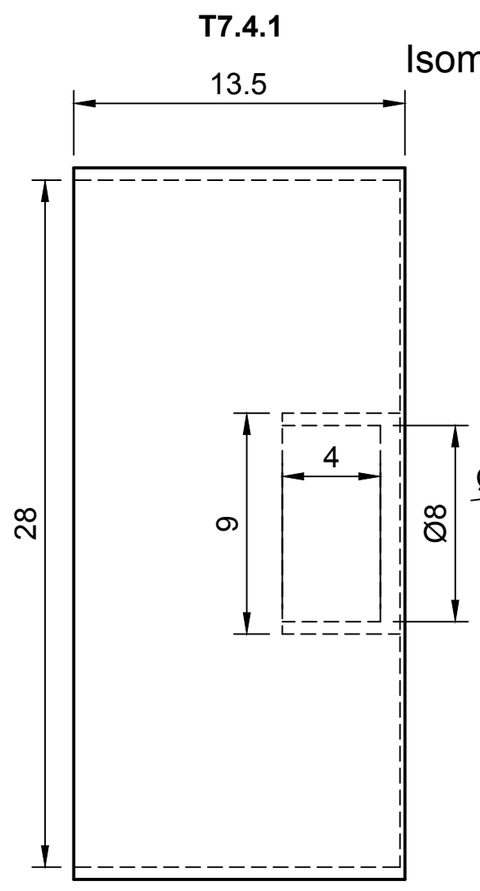
FRONT WHEEL

Material: Ketron PEEK

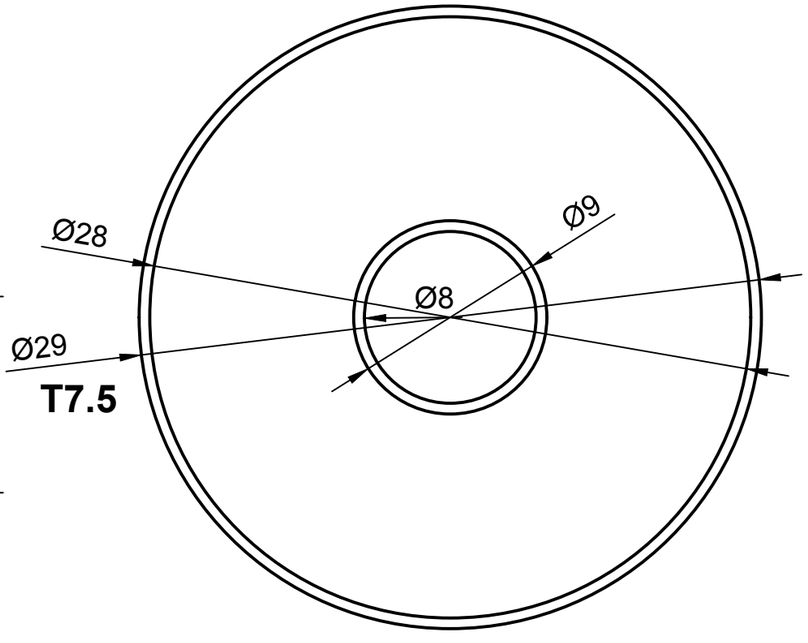
Chosen as it can be CNC machined which results in concentric and uniform wheels



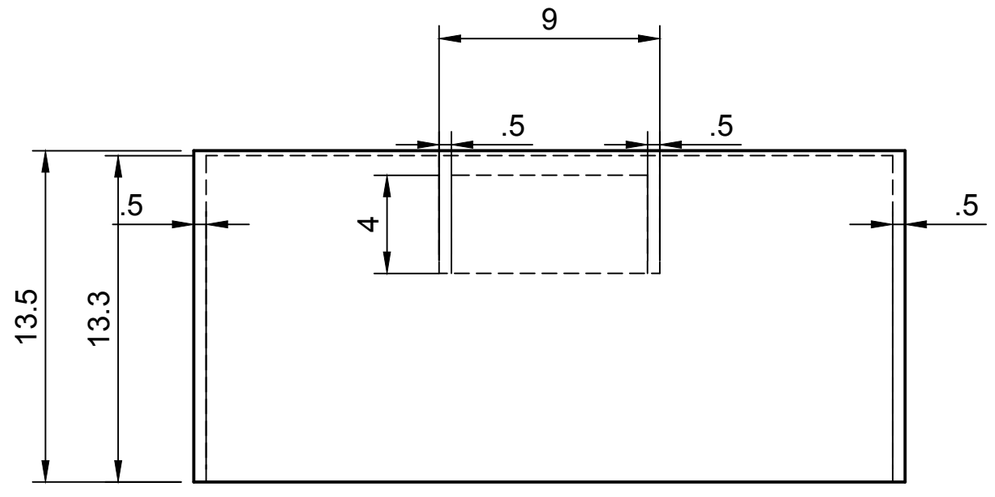
Isometric View of Front Wheel



Side View



Rear View

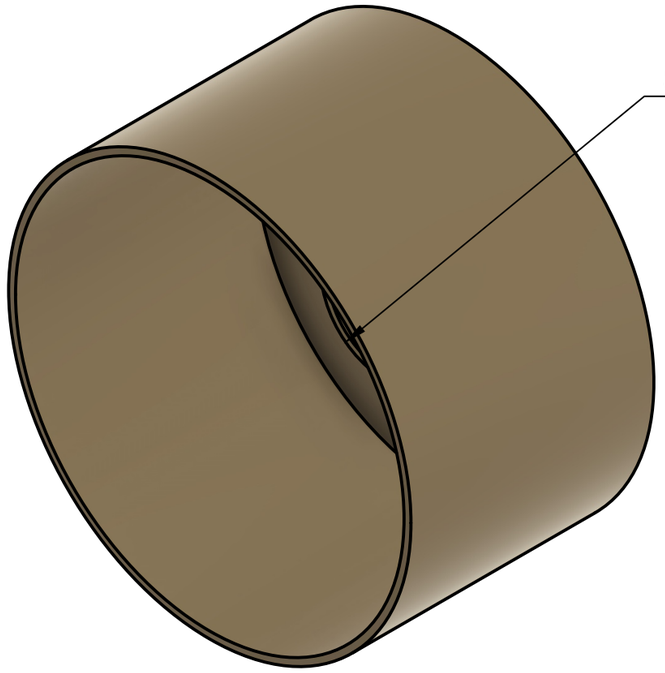


Front view

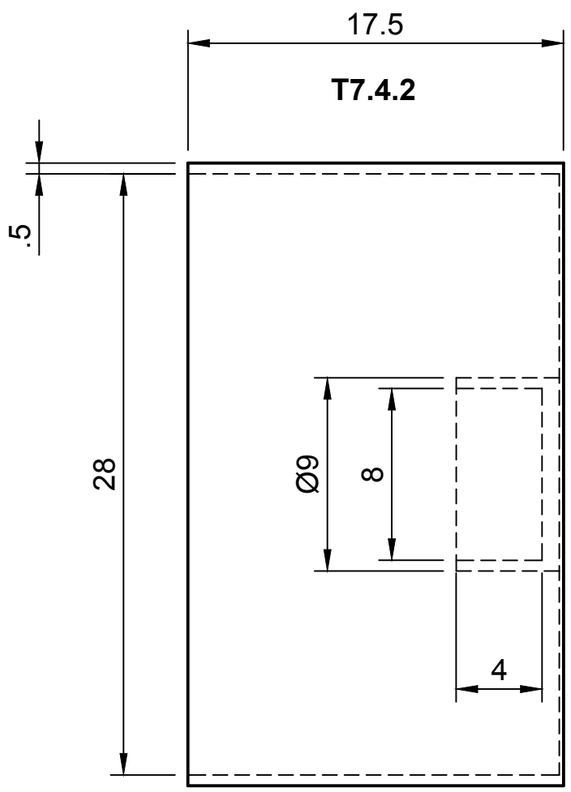
REAR WHEEL

Material: Ketron PEEK

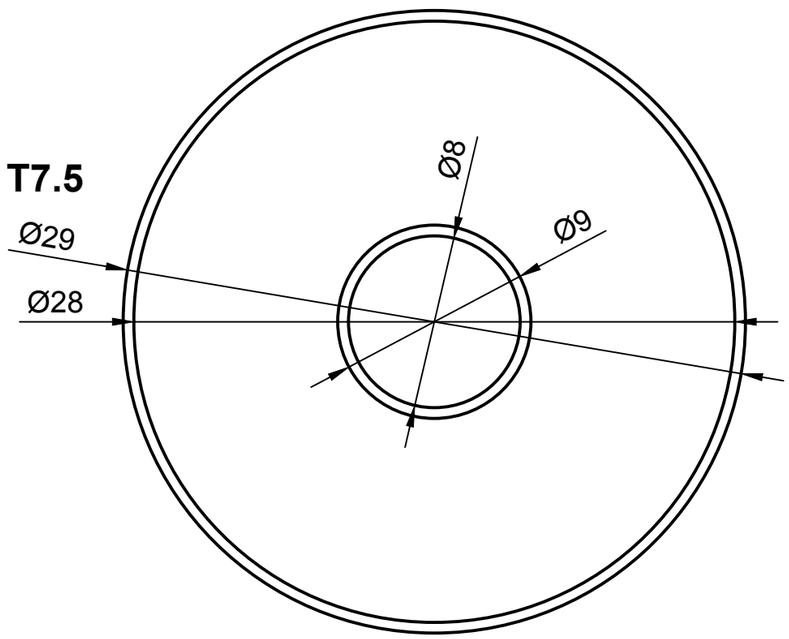
Chosen as it can be CNC machined which results in concentric and uniform wheels



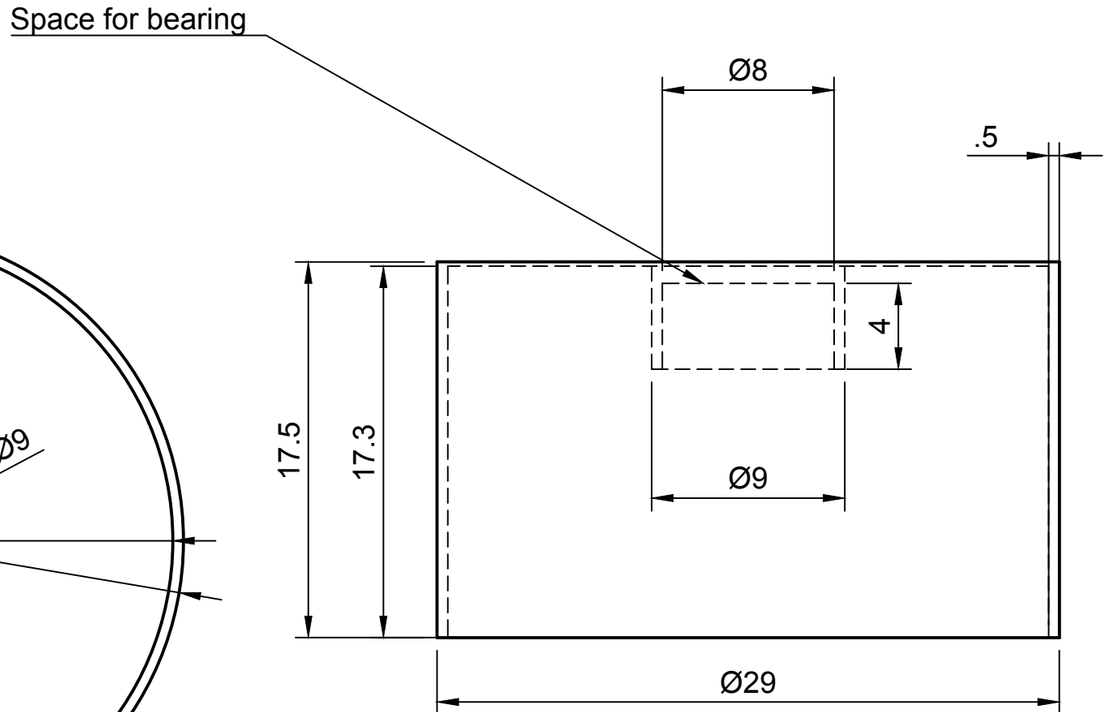
Isometric View of Rear Wheel



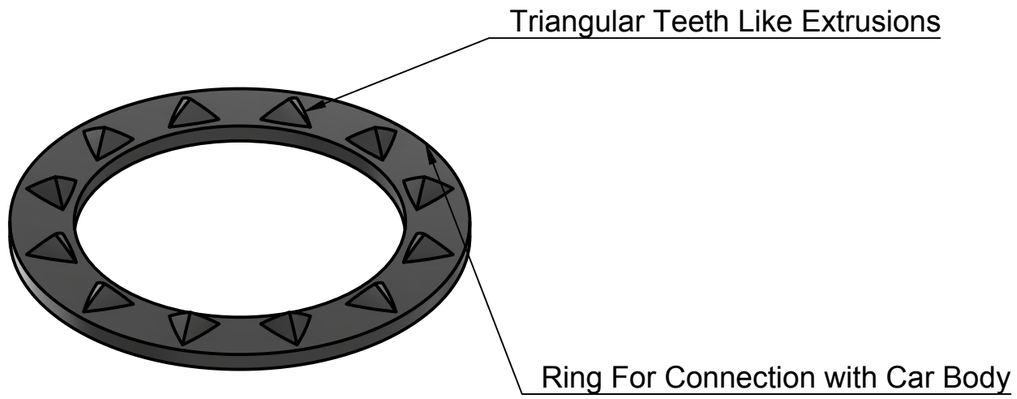
Side View



Rear view



Front View

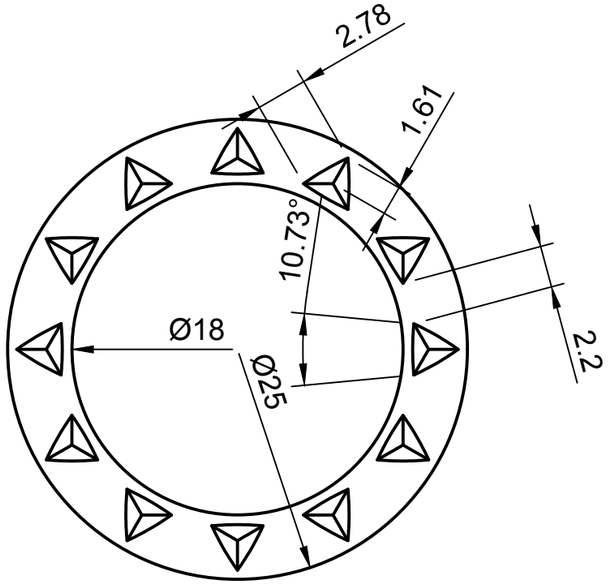


Component Identification

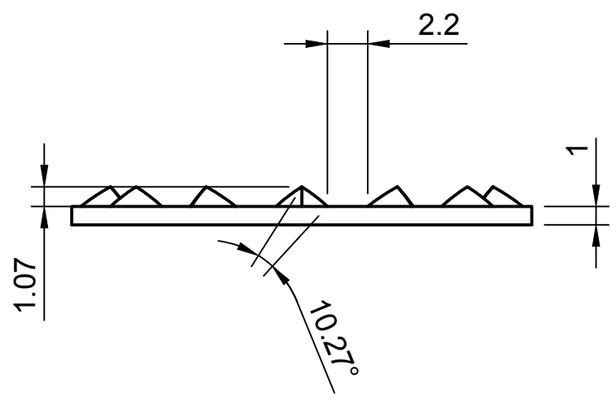
TEETH

Material Chosen: Carbon Fibre

Chosen due to its high strength to weight ration



Top View



Side View

Important Dimensions for Manufacturing